EASY ALGEBRA

Linear functions

08

8 questions · ~12 min

SAT QUESTION BANK 2026-05-29

Q1 ALGEBRA

The length, y, of a white whale was $162 \text{ centimeters } \text{cm}$ when it was born and increased an average of 4.8 cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in cm, of the whale?

A. $y = 162x$

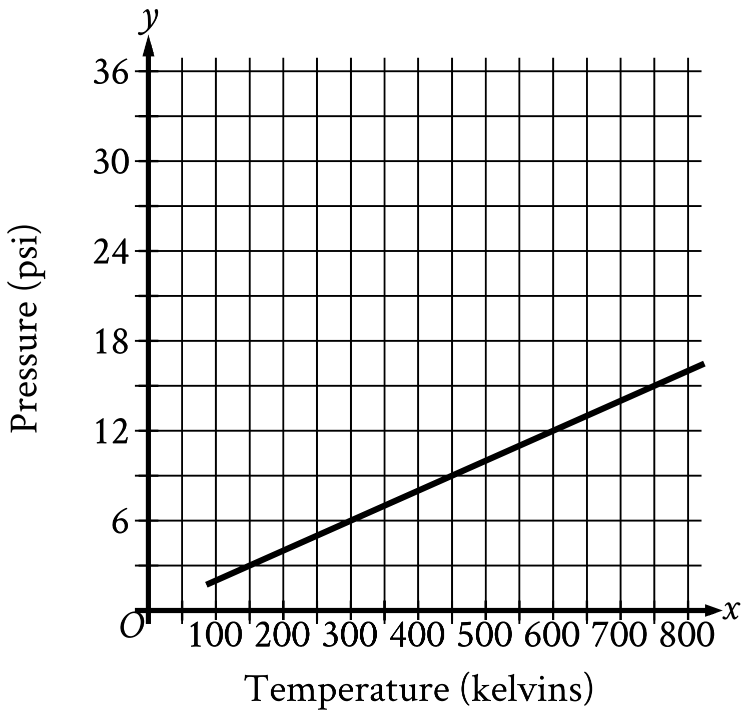
B. $y = 162x + 162$

C. $y = 4.8x + 4.8$

D. $y = 4.8x + 162$

Q2 ALGEBRA

Argon is placed inside a container with a constant volume. The graph shows the estimated pressure y, in pounds per square inch psi, of the argon when its temperature is x kelvins.



- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.8).
- The line passes through the following points:
  - (300 comma 6)
  - (450 comma 9)
  - (600 comma 12)

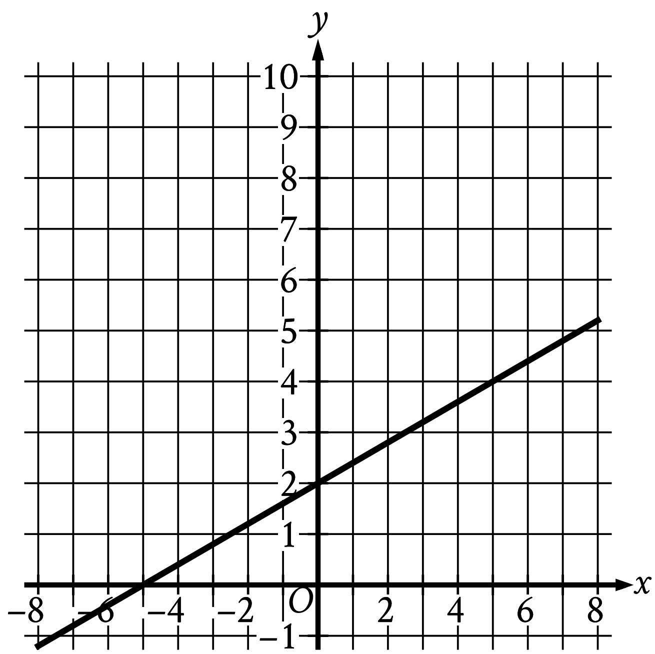
What is the estimated pressure of the argon, in psi, when the temperature is 600 kelvins?

A. 6

B. 12

C. 300

D. 600



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 5 comma 0)
  - (0 comma 2)

The graph of the linear function f is shown. What is the y-intercept of the graph of $y = fx$?

A. -5, 0

B. 2, 0

C. 0, 2

D. 0, - 5

Q4 ALGEBRA

On January 1, 2015, a city's minimum hourly wage was $9.25. It will increase by $0.50 on the first day of the year for the next 5 years. Which of the following functions best models the minimum hourly wage, in dollars, x years after January 1, 2015, where $x = 1, 2, 3, 4, 5$?

A. $f(x) = 9.25 - 0.50x$

B. $f(x) = 9.25x - 0.50$

C. $f(x) = 9.25 + 0.50x$

D. $f(x) = 9.25x + 0.50$

Q5 ALGEBRA

The function f is defined by $f x = \frac{1}{2}x + 6$. What is the value of $f 4$?

A. 20

B. 12

C. 10

D. 5

Q6

ALGEBRA

The function h is defined by $hx = 3x$ - 7. What is the value of h-2?

A. -13

B. -10

C. 10

D. 13

Q7 ALGEBRA

To repair a refrigerator, a technician charges $ 60 per hour for labor plus $ 120 for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

A. $f x = x + 120$

B. $f x = 60x$

C. $f x = 60x + 120$

D. $f x = 60x - 120$

Q8

ALGEBRA

For the linear function f, $f0 = 17$ and $f1 = 17$. Which equation defines f?

A. $fx = \frac{1}{17}$

B. $fx = 1$

C. $fx = 17$

D. $fx = 34$